



Assignment-5: String in Python

♦ String Basics

1. **Filter vowels and consonants** from the string `"How are you sir"`.
 2. **Count vowels and consonants** in the string `"How are you sir"`.
 3. **Reverse the string** `"How are you sir"`.
 4. **Convert lowercase letters to uppercase** in the string `"How are you sir"`.
 5. **Remove duplicate letters** from the string `"this is python programming place"`.
 6. **Search for a specific character** in the string `"this is python programming place"`.
 7. **Find the greatest and smallest characters** from the string `"venugopaliyer"`.
 8. **Count the total occurrences of a specific letter** in the string `"this is python programming place"`.
 9. **Replace "python" with "javascript"** in the string `"python developer python engineer python holder"`.
 10. **Print alternate letters** from the string `"How are you sir"`.
 11. **Convert the string** `"qwertyuiopasdfghjklzxcvbnm"` to `"abcdefghijklmnopqrstuvwxyz"`.
-

Extra String Questions

12. **Check if the string is a palindrome** (e.g., "madam" → Palindrome, "hello" → Not palindrome).
13. **Count spaces, digits, alphabets, and special characters** in "Python 3.9 is awesome!!".
14. **Find the longest word** in the string "Python programming is interesting".
15. **Capitalize the first letter of each word** in "welcome to python world".
16. **Remove all spaces** from "How are you sir".
17. **Check if all characters in the string are unique** (e.g., "abcde" → True, "hello" → False).
18. **Sort characters alphabetically** in "programming" → "aggimnopr".
19. **Swap cases of all letters** in "Python Is Fun" → "pYTHON iS fUN".
20. **Find frequency of each character** in "banana" → { 'b':1, 'a':3, 'n':2 }.
21. **Remove vowels** from "How are you sir" → "Hw r y sr".
22. **Check if a substring exists** in "Python programming" (e.g., "thon" → Found).
23. **Print words in reverse order** in "How are you sir" → "sir you are How".
24. **Count words** in the string "This is a python assignment".
25. **Find the ASCII value** of each character in "ABcd".

26. **Convert a string into a list of words** using `"split()"` (e.g., `"Python is fun"` → `["Python", "is", "fun"]`).
 27. **Join a list of words into a string** using `"join()"` (e.g., `["Python", "is", "fun"]` → `"Python is fun"`).
 28. **Find the first non-repeating character** in `"swiss"` → `"w"`.
 29. **Check if two strings are anagrams** (e.g., `"listen"` and `"silent"` → Anagrams).
 30. **Replace all spaces with hyphens (-)** in `"Python is easy to learn"` → `"Python-is-easy-to-learn"`.
-

Substring Related Questions

31. **Extract a substring** from `"Python Programming"` → from index 0 to 6 should give `"Python"`.
32. **Check if one string is a substring of another** (e.g., `"gram"` is a substring of `"Programming"`).
33. **Find all occurrences of a substring** in `"This is Python and Python is fun"` → Substring `"Python"`.
34. **Replace a substring** in `"I like Python"` → Replace `"Python"` with `"Java"`.
35. **Remove a substring** from `"HelloWorld"` → Remove `"World"` → `"Hello"`.
36. **Count occurrences of a substring** in `"banana"` → Substring `"ana"` appears 2 times.

37. **Check if a string starts with a substring** (e.g., "Python is easy" starts with "Python").
38. **Check if a string ends with a substring** (e.g., "Learn coding" ends with "coding").
39. **Split a string based on a substring** (e.g., "apple,banana,grapes" → Split by "," → ["apple", "banana", "grapes"]).
40. **Find the index of the first occurrence of a substring** in "Programming is great" → Substring "is" → Index 12.
41. **Find the index of the last occurrence of a substring** in "Programming in Python Programming" → Substring "Programming".
42. **Extract substring after a specific word** (e.g., "Welcome to Python World" → substring after "to" → "Python World").
43. **Extract substring before a specific word** (e.g., "Welcome to Python World" → substring before "Python" → "Welcome to").
44. **Check if two strings are rotations (cyclic substrings)** of each other (e.g., "abcd" and "cdab" → Rotations).
45. **Find the longest common substring** between two strings (e.g., "abcdxyz" and "xyzabcd" → Longest common substring = "abcd").